CENG 252: Chemical Reaction Engineering

Practice Midterm 1 (Lectures 1–5)

Time: 1 hour 20 minutes • Total: 100 points

Instructions

- Closed book, closed notes, **no cheat sheet**. Only a simple scientific calculator.
- Show all work: write the mole balance, rate law, stoichiometry (and energy balance / pressure drop where needed) *before* plugging in numbers. Partial credit is given for correct setup.
- State every assumption. Box your final answers and include units.
- Budget your time: about 0.8 minutes per point (16 minutes for a 20-point problem).

Problem	Topic	Points	Score
1	Short answers (rate laws, Arrhenius, stoichiometry)	15	
2	Levenspiel plot and reactor arrangement	20	
3	Gas-phase reversible reaction: equilibrium and CSTR	20	
4	Packed bed with pressure drop	25	
5	Adiabatic liquid-phase reactor	20	
	Total	**100**	

Constants: $R = 8.314$ J/(mol·K) $= 1.987$ cal/(mol·K) $= 0.08206$ dm^3·atm/(mol·K). 1 dm^3 = 1 L.

Problem 1: Short answers [15 pts]

(a) **[3]** For the elementary reaction $2\,\mathrm{A} + \mathrm{B} \longrightarrow 3\,\mathrm{C}$, the rate of disappearance of A at some point in a reactor is $-r_A = 6$ mol/(dm^3·s). What are $-r_B$ and r_C at that point?

(b) **[3]** The rate law is $-r_A = k\,C_A^2 C_B$. Give the units of k in terms of mol, dm^3, and s.

(c) **[3]** The rate constant of a reaction doubles when the temperature is raised from 300 K to 310 K. Find the activation energy E (in cal/mol). By what factor does k increase from 300 K to 320 K? (Is it exactly 4?)

(d) **[3]** An irreversible reaction is run isothermally to the same conversion in a CSTR and in a PFR with the same feed. Compare V_{CSTR} and V_{PFR} (larger, smaller, or equal) for (i) a zero-order reaction and (ii) a first-order reaction. Justify each answer with a sketch of the Levenspiel plot.

(e) **[3]** Pure gaseous A is fed to an isothermal PFR where $\mathrm{A} \longrightarrow 2\,\mathrm{B}$ takes place with negligible pressure drop. At $X_A = 0.5$, what is v/v_0, and what is C_A in terms of C_{A0}?

Problem 2: Levenspiel plot and reactor arrangement [20 pts]

An exothermic liquid-phase reaction A $\longrightarrow$ B is carried out **adiabatically**. The following data were measured with $F_{A0} = 1$ mol/min:

X_A	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
$F_{A0}/(-r_A)$ (dm^3)	100	70	50	36	27	22	20	24	40

(a) **[3]** Sketch the Levenspiel plot. In one or two sentences, explain physically why $F_{A0}/(-r_A)$ first *decreases* and then *increases* with X_A.

(b) **[4]** What single CSTR volume is needed to reach $X_A = 0.8$?

(c) **[6]** What single PFR volume is needed to reach $X_A = 0.8$? Use Simpson's rule with all nine data points.

(d) **[7]** You may use **one CSTR and one PFR in series** (either order) to reach $X_A = 0.8$. Which arrangement and which intermediate conversion give the *smallest total volume*? Compute that total volume and compare it with (b) and (c). Shade the corresponding areas on your sketch from (a).

Problem 3: Gas-phase reversible reaction [20 pts]

The elementary, reversible gas-phase reaction

$$\mathrm{A} \rightleftharpoons 2\,\mathrm{B}, \qquad -r_A = k_A \left[C_A - \frac{C_B^2}{K_C} \right]$$

is carried out isothermally at 340 K and constant total pressure (no pressure drop). The feed is **50 mol% A and 50 mol% inert I** at a total pressure of 2 atm. At 340 K, $K_C = 0.1$ mol/dm^3 and $k_A = 0.5$ min^{-1}.

(a) **[5]** Set up the stoichiometric table for a **flow** system (species A, B, I; columns: feed, change, exit). Find C_{A0}, δ, and ε, and write C_A and C_B as functions of X.

(b) **[5]** Calculate the equilibrium conversion X_e in the flow reactor.

(c) **[4]** The same feed (same initial C_{A0}) is instead placed in a **constant-volume batch reactor**. Calculate X_e for this case. Explain in one sentence why it differs from (b).

(d) **[6]** The flow reactor is a CSTR with $v_0 = 50$ dm^3/min. What CSTR volume is needed to reach $X = 0.8\,X_e$ (with X_e from part (b))? Also give the space time τ.

Problem 4: Packed-bed reactor with pressure drop [25 pts]

The gas-phase catalytic isomerization A $\longrightarrow$ B is first order in A and is carried out isothermally in a packed-bed reactor (PBR) containing $W = 250$ kg of catalyst:

$$-r'_A = k'C_A, \qquad k' = 0.06\ \frac{\text{dm}^3}{\text{kg cat} \cdot \text{min}}.$$

Pure A is fed at $v_0 = 10$ dm^3/min. The flow through the bed is **turbulent**, and the pressure-drop parameter is $\alpha = 0.002$ kg^{-1}.

(a) **[7]** Starting from the PBR mole balance and the Ergun equation in the form $\frac{dy}{dW} = -\frac{\alpha}{2y}(1+\varepsilon X)\frac{T}{T_0}$, show that $y = (1-\alpha W)^{1/2}$, and derive an analytic expression for X as a function of W.

(b) **[5]** Calculate the exit conversion and the exit pressure ratio $y = P/P_0$. What would the conversion be if pressure drop were neglected?

(c) **[4]** The engineer proposes adding more catalyst to the same bed (same particles, same flow) to raise the conversion. What is the largest catalyst weight the bed can hold before the pressure falls to zero, and what is the **maximum** conversion this design could ever reach?

(d) **[4]** Management wants to **double the throughput**: F_{A0} and v_0 are doubled, while the inlet pressure, temperature, and composition (and the reactor and catalyst) stay the same. By what factor does α change? Show that the original 250 kg bed can no longer be operated.

(e) **[5]** To fix the problem in (d), larger catalyst particles are used. By what factor must D_p be increased so that the exit pressure y is the same as in (b)? What is the exit conversion then? (Assume that k', the bulk density, and the void fraction do not change.) Comment on the trade-off.

Problem 5: Adiabatic liquid-phase reactor [20 pts]

The irreversible, elementary liquid-phase reaction

$$\mathrm{A} + \mathrm{B} \longrightarrow \mathrm{C}, \qquad -r_A = k\, C_A C_B$$

is carried out **adiabatically**. The feed contains equimolar A and B with $C_{A0} = C_{B0} = 1.0$ mol/dm^3, plus an inert solvent I with $\Theta_I = 4$. The feed enters at $T_0 = 300$ K with $\upsilon_0 = 10$ dm^3/min.

Given

Heat of reaction:	$\Delta H^\circ_{\mathrm{rxn}}(T_R = 298\ \mathrm{K}) = -10{,}000$ cal/mol A
Heat capacities:	$\widetilde{C}_{pA} = \widetilde{C}_{pB} = 20,\ \widetilde{C}_{pC} = 30,\ \widetilde{C}_{pI} = 18$ cal/(mol·K)
Kinetics:	$k = 0.01$ dm^3/(mol·min) at 300 K, $\quad E = 12{,}000$ cal/mol

(a) **[5]** Starting from the adiabatic energy balance, derive T as an explicit function of X. Do **not** assume $\Delta\widetilde{C}_p = 0$. Evaluate T at $X = 0.8$.

(b) **[4]** What CSTR volume is needed to reach $X = 0.8$?

(c) **[7]** What PFR volume is needed to reach $X = 0.8$? Use Simpson's rule with five points ($X = 0$, 0.2, 0.4, 0.6, 0.8). Organize your work in a table of X, T, k, and $F_{A0}/(-r_A)$.

(d) **[4]** Which reactor in (b) and (c) is smaller? Use the shape of your $F_{A0}/(-r_A)$ column to explain why this is *not* the usual result for a positive-order reaction. Also, by how many kelvin would you have mispredicted the exit temperature at $X = 0.8$ if you had wrongly set $\Delta\widehat{C}_p = 0$?

Possibly useful integrals (cover this box on your first attempt: the real exam may not provide them)

$$\int_0^X \frac{dX}{1-X} = \ln\frac{1}{1-X}, \qquad \int_0^X \frac{dX}{(1-X)^2} = \frac{X}{1-X}$$

$$\int_0^X \frac{1+\varepsilon X}{1-X}\,dX = (1+\varepsilon)\ln\frac{1}{1-X} - \varepsilon X, \qquad \int_0^W (1-\alpha W)^{1/2}\,dW = \frac{2}{3\alpha}\left[1-(1-\alpha W)^{3/2}\right]$$